

Linear Algebra and Calculus Lecture 12

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Dimension

In this lecture we define the notion of dimension of a vector space.

Definition. If a vector space V is spanned by a finite set, then V is said to be **finite-dimensional**, and the **dimension** of V , written as $\dim V$, is the number of vectors in a basis for V . The dimension of the zero vector space $\{0\}$ is defined to be zero. If V is not spanned by a finite set, then V is said to be **infinite-dimensional**.

Example. Some examples:

- The standard basis for $\mathbb{R}^n$ contains n vectors, so $\dim \mathbb{R}^n = n$.
- The standard polynomial basis $\{1, t, t^2\}$ shows that $\dim \mathbb{P}_2 = 3$.
- In general, $\dim \mathbb{P}_n = n + 1$.

Theorem. Let H be a subspace of a finite-dimensional vector space V . Any linearly independent set in H can be expanded, if necessary, to a basis for H . Also, H is finite-dimensional and

$$\dim H \leq \dim V.$$

Theorem. Let V be a p -dimensional vector space, $p \geq 1$. Any linearly independent set of exactly p elements in V is automatically a basis for V . Any set of exactly p elements that spans V is automatically a basis for V .

Definition. The **rank** of an $m \times n$ matrix A is the dimension of the column space and the **nullity** of A is the dimension of the null space.

Proposition. The rank of an $m \times n$ matrix A is the number of pivot columns and the nullity of A is the number of free variables.

Theorem. (The Rank Theorem) The dimensions of the column space and the null space of an $m \times n$ matrix A satisfy the equation

$$\text{rank } A + \text{nullity } A = \text{number of columns in } A$$

Exercise. If A is a 7×9 matrix with nullity 2, what is the rank of A ? Could a 6×9 matrix have nullity 2?

Theorem. (*The Invertible Matrix Theorem DLC*) We can now add the following statements to the invertible matrix theorem that we saw in previous lectures:

(m) The columns of A form a basis of $\mathbb{R}^n$.

(n) $\text{Col } A = \mathbb{R}^n$.

(o) $\text{rank } A = n$.

(p) $\text{nullity } A = 0$.

(q) $\text{Nul } A = \{0\}$.

Coordinate Systems

Theorem. (*The Unique Representation Theorem*) Let $\mathcal{B} = \{b_1, \dots, b_n\}$ be a basis for a vector space V . Then for each $x \in V$, there exists a unique set of scalars $c_1, \dots, c_n$ such that

$$x = c_1 b_1 + \dots + c_n b_n.$$

Definition. Suppose $\mathcal{B} = \{b_1, \dots, b_n\}$ is a basis for a vector space V and x is in V . The **coordinates of x relative to the basis $\mathcal{B}$** (or the **$\mathcal{B}$ -coordinates of x**) are the weights $c_1, \dots, c_n$ such that $x = c_1 b_1 + \dots + c_n b_n$.

If $c_1, \dots, c_n$ are the $\mathcal{B}$ -coordinates of x , then the vector in $\mathbb{R}^n$

$$[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

is the coordinate vector of x (relative to $\mathcal{B}$), or the $\mathcal{B}$ -coordinate vector of x . The mapping $x \mapsto [x]_{\mathcal{B}}$ is the coordinate mapping (determined by $\mathcal{B}$).

Example. Let $\mathbf{b}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $x = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$, and $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$. Find the coordinate vector $[x]_{\mathcal{B}}$ of x relative to $\mathcal{B}$.

The $\mathcal{B}$ -coordinates c_1, c_2 of x satisfy

$$c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

That is,

$$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

This equation can be solved by row operations on an augmented matrix or by multiplying the vector x by the inverse of the matrix. In any case, the solution is $c_1 = 3$, $c_2 = 2$. Thus $x = 3\mathbf{b}_1 + 2\mathbf{b}_2$, and

$$[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

The matrix $P_{\mathcal{B}}$ is the **change-of-coordinates matrix** from $\mathcal{B}$ to the standard basis in $\mathbb{R}^n$, with

$$P_{\mathcal{B}} = \begin{bmatrix} b_1 & \cdots & b_n \end{bmatrix},$$

so that $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$. Since the matrix is made out of vectors of a basis, it is invertible, so $P_{\mathcal{B}}^{-1}$ converts x into its $\mathcal{B}$ -coordinate vector: $P_{\mathcal{B}}^{-1}x = [x]_{\mathcal{B}}$.

Example. Consider two bases $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2\}$ for a vector space V , such that

$$\mathbf{b}_1 = 4\mathbf{c}_1 + \mathbf{c}_2 \quad \text{and} \quad \mathbf{b}_2 = -6\mathbf{c}_1 + \mathbf{c}_2$$

Suppose

$$\mathbf{x} = 3\mathbf{b}_1 + \mathbf{b}_2.$$

That is, suppose $[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Let's find $[\mathbf{x}]_{\mathcal{C}}$. Apply the coordinate mapping determined by $\mathcal{C}$ to $\mathbf{x}$ in the expression for x . Since the coordinate mapping is a linear transformation,

$$[\mathbf{x}]_{\mathcal{C}} = [3\mathbf{b}_1 + \mathbf{b}_2]_{\mathcal{C}} = 3[\mathbf{b}_1]_{\mathcal{C}} + [\mathbf{b}_2]_{\mathcal{C}}$$

We can write this vector equation as a matrix equation, using the vectors in the linear combination as the columns of a matrix:

$$[\mathbf{x}]_{\mathcal{C}} = \begin{bmatrix} [\mathbf{b}_1]_{\mathcal{C}} & [\mathbf{b}_2]_{\mathcal{C}} \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

This formula gives $[\mathbf{x}]_{\mathcal{C}}$, once we know the columns of the matrix. From the expressions of $\mathbf{b}_1$ and $\mathbf{b}_2$,

$$[\mathbf{b}_1]_{\mathcal{C}} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \quad \text{and} \quad [\mathbf{b}_2]_{\mathcal{C}} = \begin{bmatrix} -6 \\ 1 \end{bmatrix}$$

Thus,

$$[\mathbf{x}]_{\mathcal{C}} = \begin{bmatrix} 4 & -6 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}.$$

Theorem. Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and $\mathcal{C} = \{\mathbf{c}_1, \dots, \mathbf{c}_n\}$ be bases of a vector space V . Then there is a unique $n \times n$ matrix $P_{\mathcal{C} \leftarrow \mathcal{B}}$ such that

$$[\mathbf{x}]_{\mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}} [\mathbf{x}]_{\mathcal{B}}$$

The columns of $P_{\mathcal{C} \leftarrow \mathcal{B}}$ are the $\mathcal{C}$ -coordinate vectors of the vectors in the basis $\mathcal{B}$. That is,

$$P_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} [\mathbf{b}_1]_{\mathcal{C}} & [\mathbf{b}_2]_{\mathcal{C}} & \cdots & [\mathbf{b}_n]_{\mathcal{C}} \end{bmatrix}$$

Using all the previous results we see that $(P_{\mathcal{C} \leftarrow \mathcal{B}})^{-1} = P_{\mathcal{B} \leftarrow \mathcal{C}}$

Example. Let $\mathbf{b}_1 = \begin{bmatrix} -9 \\ 1 \end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix} -5 \\ -1 \end{bmatrix}$, $\mathbf{c}_1 = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$, $\mathbf{c}_2 = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$, and consider the bases for $\mathbb{R}^2$ given by $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2\}$. Find the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$.

The matrix ${}_{\mathcal{C} \leftarrow \mathcal{B}}^P$ involves the $\mathcal{C}$ -coordinate vectors of $\mathbf{b}_1$ and $\mathbf{b}_2$. Let

$$[\mathbf{b}_1]_{\mathcal{C}} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{and} \quad [\mathbf{b}_2]_{\mathcal{C}} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}.$$

Then, by definition,

$$[\mathbf{c}_1 \quad \mathbf{c}_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{b}_1 \quad \text{and} \quad [\mathbf{c}_1 \quad \mathbf{c}_2] \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \mathbf{b}_2.$$

To solve both systems simultaneously, augment the coefficient matrix with $\mathbf{b}_1$ and $\mathbf{b}_2$, and row reduce:

$$[\mathbf{c}_1 \quad \mathbf{c}_2 \quad \mathbf{b}_1 \quad \mathbf{b}_2] = \begin{bmatrix} 1 & 3 & -9 & -5 \\ -4 & -5 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 6 & 4 \\ 0 & 1 & -5 & -3 \end{bmatrix}$$

Thus,

$$[\mathbf{b}_1]_{\mathcal{C}} = \begin{bmatrix} 6 \\ -5 \end{bmatrix} \quad \text{and} \quad [\mathbf{b}_2]_{\mathcal{C}} = \begin{bmatrix} 4 \\ -3 \end{bmatrix}.$$

The desired change-of-coordinates matrix is therefore

$${}_{\mathcal{C} \leftarrow \mathcal{B}}^P = [[\mathbf{b}_1]_{\mathcal{C}} \quad [\mathbf{b}_2]_{\mathcal{C}}] = \begin{bmatrix} 6 & 4 \\ -5 & -3 \end{bmatrix}.$$

From this we can deduce that

$$[\mathbf{c}_1 \quad \mathbf{c}_2 \quad | \quad \mathbf{b}_1 \quad \mathbf{b}_2] \sim \left[I \quad | \quad {}_{\mathcal{C} \leftarrow \mathcal{B}}^P \right].$$